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● *Original Contribution*

CONSERVATIVE TREATMENT OF UVEAL MELANOMA: PROBABILITY OF EYE RETENTION AFTER PROTON TREATMENT

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Enucleation was performed after proton treatment in 57 of 1006 (5.7%) uveal melanoma patients treated with proton beam therapy at the Harvard Cyclotron Laboratory between July 1975 and December 31, 1986. Only 2% of 99 patients with small tumors and 4% of 566 patients with intermediate size tumors underwent enucleation after treatment; 10% of 341 patients with large tumors lost the treated eye. No eyes were removed after 52 months, with 89% of enucleations performed during the first 36 months after treatment. Eye retention rates at 60 months were 89.1 ± 3.0% for the entire group, and 97 ± 3.7%, 92.7 ± 3.1%, and 78.3 ± 7.0% in patients with small, intermediate, and large tumors, respectively. Significantly greater enucleation rates were observed in patients with large tumors than in those with intermediate tumors (p = <.0001), in patients with tumor height > 8 mm relative to those with tumors ≤ 8 mm, p = (<.0001), with tumor diameter > 16 mm compared to ≤16 mm, (p = <.0001), and with tumor involvement of the ciliary body compared to involvement of the choroid only (p = <.0001). Possible strategies to decrease the likelihood of enucleation in patients at apparently increased risk of losing the eye after conservative therapy, that is, those with large tumors involving the ciliary body, might include a lower total dose, a more protracted treatment course, or a lower radiation dose and adjuvant treatment with chemotherapy and/or immunotherapy, with hyperthermia, or with other radiation sensitizers.

Conservative treatment, Eye, Ocular melanoma, Uveal melanoma, Proton beam therapy, Enucleation.

INTRODUCTION

Recent reports have documented that vision is being preserved in most conservatively treated uveal melanoma patients.[2,3,5,10,17,18,21] Survival does not appear to have been compromised by allowing the treated eye to remain in place rather than resorting to radical treatment, that is, enucleation, immediately after diagnosis or documentation growth.[1,13,15,20] Some eyes, however, have had to be removed after treatment, due to complications or tumor growth in patients treated conservatively with external beam charged particle therapy[10,18] or radionuclide plaque therapy.[5,13,16,17] This study was undertaken to determine the rate of eye loss after conservative treatment with proton beam radiation, and to identify patient groups with an increased risk of enucleation after such treatment, using univariate statistical analysis.

METHODS AND MATERIALS

Between 1975 and December, 1986, 1007 uveal melanomas in 1006 patients were treated with the 160 MeV proton beam at the Harvard Cyclotron Laboratory, in a collaborative effort between the staff of that Laboratory, the Retina Service of the Massachusetts Eye and Ear Infirmary, and the Radiation Medicine Service of the Massachusetts General Hospital. Five hundred twenty two

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